AF^{2}DN: Attention-Guided Frequency Feature Decomposition Network for Hyperspectral and LiDAR Data Classification

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***Abstract*—Transformers have gained significant attention in the multimodal remote sensing fusion due to their strong global context modeling capability. Although Transformer-based methods excel at processing high-dimensional spectral sequences and joint spatial–spectral information, most current research remains focused on the spatial domain. Consequently, the exploration of frequency-domain features—particularly implicit frequency representations—is often neglected. Moreover, efficiently fusing multimodal data features while emphasizing more discriminative information remains a challenging task. To address these challenges, this article proposes an attention-guided frequency feature decomposition network (AF^{2}DN) for hyperspectral and light detection and ranging (LiDAR) data classification. First, a Transformer-based frequency feature decomposition (TFFD) method is proposed, employing the window attention to capture distinct directional frequency components from multimodal remote sensing data. Through this approach, low-frequency components are utilized to characterize global structural information, while various high-frequency components are employed to extract diverse texture and edge features. Second, an attention frequency modulation (AFM) module is developed, incorporating a weight learning matrix in the frequency domain. This matrix is designed to selectively amplify and suppress different frequency components, thereby reducing data redundancy resulting from frequency feature decomposition. Finally, an adaptive multimodal same-frequency feature fusion (AMSF^3) module is designed to achieve cross-modal feature integration at identical frequency bands. Extensive experiments are conducted on three benchmark datasets, and the results demonstrate that the proposed framework outperforms existing state-of-the-art methods while exhibiting stronger adaptability in complex environments.**

***Index Terms*—Fast Fourier transform (FFT), feature fusion, frequency feature, hyperspectral imagery (HSI), light detection and ranging (LiDAR).**

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I. Introduction

REMOTE sensing data, captured by satellites, aircraft, and unmanned aerial vehicles (UAVs), characterizes ground objects through electromagnetic properties such as reflection, radiation, and emission. By analyzing spectral, textural, and morphological features, classification algorithms assign pixels or regions to specific land cover categories [1], [2], [3] [4], enabling effective identification of surface characteristics to support decision-making, resource management, and environmental conservation. Data from different sensors offer complementary advantages: hyperspectral imagery (HSI) captures narrow spectral bands for material identification, while light detection and ranging (LiDAR) provides accurate 3-D surface characterization. However, single-modality approaches face limitations—HSI is susceptible to environmental interference and spatial resolution constraints, whereas LiDAR lacks spectral information. Integrating HSI and LiDAR data effectively mitigates these issues and significantly improves the classification performance.

During the initial phase of multimodal remote sensing research, investigators developed fundamental fusion methodologies, including decision-level and feature-level integration approaches, by combining complementary data types. Decision-level fusion is a method of fusion after classification, which has a relatively light computational burden, but cannot fully explore the deep-level associations between different modalities. The feature-level fusion combines raw data characteristics before classification, offering enhanced information integration at the risk of dimensionality-related challenges. Chatzis et al. [5] proposed the use of clustering algorithms for decision-level fusion and proposed the fuzzy k-means algorithm and fuzzy vector quantization algorithm based on fuzzy vector distance definition. Metallinou et al. [6] used a Bayesian framework to fuse classifiers to model facial, voice, and head movement cues for emotion recognition, and the results of fusion classification significantly outperformed single classifiers. These conventional approaches typically depend on the

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prior knowledge and handcrafted feature engineering, consequently exhibiting constrained generalization performance and often failing to achieve the optimal classification accuracy results [7], [8], [9].

Deep learning has emerged as a powerful technique for multimodal remote sensing data fusion and classification, capable of automatically learning hierarchical features from raw data and demonstrating strong performance in feature extraction and classification tasks [10], [11], [12], [13]. For example, Shen et al. [14] proposed a feature-enhanced fusion network to mitigate noise caused by heterogeneous feature differences; Song et al. [15] designed a two-stream CNN based on deep hashing to provide complementary information; Zhang et al. [16] used a three-branch CNN to learn spectral, spatial, and elevation features with multilevel fusion; and Li et al. [17] employed a saliency reinforcement network to integrate spatial and spectral features through weighting, alleviating the challenge of spectral variability. However, CNN-based methods are limited by their limited receptive field. Although they can effectively extract local features, they cannot capture global contextual dependencies well.

Transformers have become widely adopted in multimodal remote sensing classification due to their strong global feature representation capabilities. Transformers differ fundamentally from CNNs by processing entire sequences simultaneously through global self-attention rather than local convolution operations. Roy et al. [18] proposed a multimodal fusion Transformer network that can learn long-term feature dependencies, which helps to better generalize the model and improve classification accuracy when processing complementary information. Ma et al. [19] used a multilayer multimodal fusion scheme for FTransUNet, which provides a robust and effective multimodal fusion backbone by integrating CNN and Vit into a unified fusion framework. Dong et al. [20] proposed the spatial context and deredundancy fusion network (SCDNet), which adaptively fuses global and local information of different scales to obtain more accurate context by developing a multiscale attention fusion module. Zhang et al. [21] used a model of a multimodal Transformer to capture specific and shared features of HSI and LiDAR data. Despite their demonstrated efficacy in extracting heterogeneous global features, existing Transformer-based methods exhibit several critical limitations. These methods predominantly focus on spatial-domain analysis, neglecting the role of frequency-domain representations. This oversight results in insufficient preservation of spectral and textural details, data redundancy, and high computational complexity. The lack of explicit frequency modeling limits their ability to capture essential structural and spectral patterns, underscoring the need for integrating frequency-domain learning for more discriminative and efficient feature representation in multimodal remote sensing.

In recent years, the frequency-domain feature learning has emerged as a key paradigm in remote sensing image analysis. While traditional methods rely on explicit mathematical transformations like the Fourier and wavelet transforms, their fixed basis functions limit the representation of complex spatial–spectral features. With advances in deep learning, implicit the frequency-domain learning has gained increasing

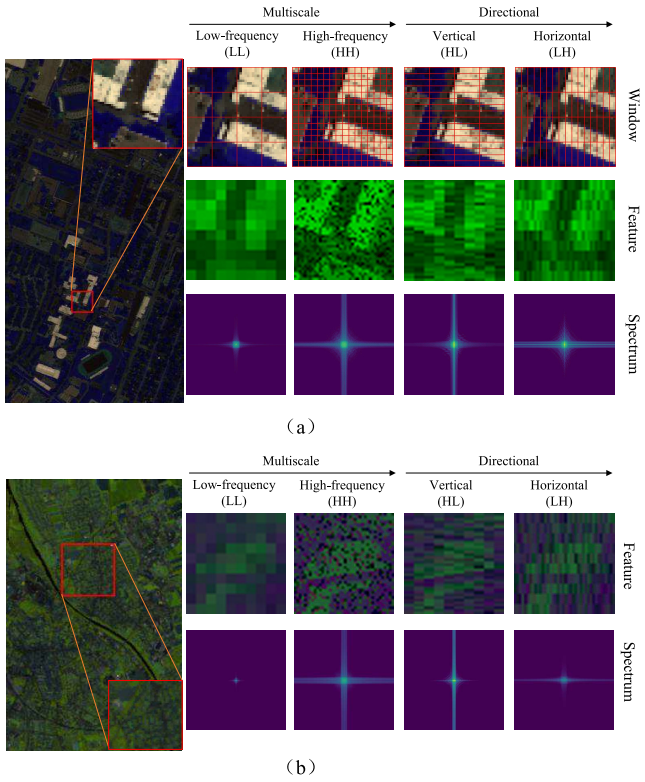


Fig. 1. Motivation of the proposed AF^{2}DN framework. (a) (Left) Randomly selected patches from the Houston2013 dataset. (Right) Window partitioning methods corresponding to different frequency-domain features, visualization results of various frequency-domain features, and their corresponding magnitude spectra. (b) (Left) Randomly selected patches from the Augsburg dataset. (Right) Visualization results of different frequency-domain features and their corresponding magnitude spectra.

attention. This approach avoids predefined transformations, instead using learnable structures—such as convolutional kernels and attention mechanisms—to adaptively decompose and reorganize implicit frequency components directly from data. Through end-to-end training, networks autonomously discover task-relevant frequency representations, demonstrating strong capabilities in joint spatial–spectral-frequency modeling.

To address the insufficient utilization of frequency-domain features in hyperspectral and LiDAR classification, we propose an attention-guided frequency feature decomposition network (AF^{2}DN). The framework autonomously learns to decompose and weight implicit frequency components in features, avoiding explicit frequency transforms [22] [23]. Its core contribution lies in the synergistic design of three specialized modules: a Transformer-based frequency feature decomposition (TFFD) module for multiscale spectral analysis, an attention frequency modulation (AFM) module that adaptively emphasizes critical frequency components via a learnable weighting matrix [24] [25], and an adaptive same-frequency feature fusion (AMSF^3) module. Specifically, as shown in Fig. 1, window attention decomposes multifrequency features by capturing anisotropic components and preserving directional details. AF^{2}DN significantly improves the implicit frequency feature modeling, enabling more precise detection of subtle variations and effective cross-modal frequency align-

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ment, with demonstrated advantages in preserving texture details, adaptive frequency weighting, and multimodal fusion.

In summary, the main contributions of AF²DN can be summarized as follows.

1) We propose a TFFD module, which for the first time combines a multihead self-attention mechanism with windows of different shapes to capture directional frequency components. This innovative design establishes long-range dependencies across frequency bands while preserving the structural information through attention-guided windowing, ultimately achieving significant enhancement in discriminating fine-grained textures and edges.
2) We propose an AFM module, which introduces a learnable frequency-domain filter matrix that dynamically suppresses redundant components. By integrating this frequency modulation (FM) layer with an attention mechanism, the module effectively enables local–global feature learning and significantly enhances the representational capacity of fused features.
3) We propose an adaptive multimodal same-frequency feature fusion (AMSF³) module that employs a coarse-to-fine weighting strategy to enhance feature consistency while effectively reducing both fusion redundancy and cross-modal discrepancies. This mechanism enables more precise extraction of discriminative features. Comprehensive experiments on three benchmark datasets demonstrate the competitive performance of our AF²DN algorithm.

The remainder of this article is organized as follows. Section I reviews related work, Section II-B presents the proposed network architecture, Section IV details experiments and analysis, and Section V concludes with future research directions.

II. Related Works

A. Transformer

Originally developed for sequence modeling in natural language processing, the Transformer model has demonstrated remarkable capabilities in image processing tasks. This success stems from its self-attention mechanism, which effectively captures long-range dependencies in parallel [26], [27], [28]. Self-attention mechanisms outperform traditional CNNs by modeling global dependencies rather than local features, proving especially effective for multimodal classification tasks involving HSI and LiDAR data integration [29].

Currently, Transformers face three key challenges in multimodal remote sensing fusion classification: inadequate local feature representation, high computational complexity, and poor multimodal alignment. Ding et al. [30] proposed a global local exchange network that combines the advantages of convolutional operators in characterizing locally relevant features and the promising ability of Transformer in learning long-range dependencies to further improve the classification performance. Wang et al. [31] introduced the Pyramid Visual Transformer (PVT) to reduce the computational cost of large feature maps by shrinking the pyramid asymptotically. To address the problem of incomplete use of category representation information due to multimodal heterogeneous features, Yang et al. [32] proposed a multimodal fusion visual Transformer framework, and the proposed multimodal cross-attention module reduces the alignment requirement between modal feature spaces during cross-modal fusion. However, most existing Transformer-based multimodal remote sensing fusion methods operate solely in the spatial domain, neglecting critical frequency-domain features that preserve edge and texture details. Conventional CNN–Transformer fusion methods primarily rely on CNNs for local spatial feature extraction, making them susceptible to noise interference and ineffective at capturing high-order frequency-domain features, ultimately compromising classification accuracy. Hyperspectral data often contains substantial redundancy, which can significantly impair training efficiency, particularly with large-scale datasets. Furthermore, CNN–Transformer fusion methods often fail to effectively model intermodal feature correlations due to limitations in CNN-based feature alignment. The frequency-domain Transformer approach effectively captures cross-modal dependencies while mitigating intermodal discrepancies, thereby simplifying and enhancing the efficiency of cross-modal modeling.

B. Feature Fusion

Multimodal feature fusion integrates complementary information from diverse sensing modalities (e.g., optical, radar, LiDAR, and hyperspectral data) to enhance data representation and boost performance in classification, detection, and segmentation tasks. Recent advances in remote sensing and computer vision demonstrate that this approach significantly improves system accuracy and robustness [33], [34], [35], [36], [37]. However, as research progresses, increasing challenges and limitations have emerged.

Conventional decision-level fusion methods independently processed each modality prior to applying deterministic combination rules, potentially disregarding latent cross-modal dependencies [38]. Feature-level fusion, another prevalent approach, aligns multimodal data before concatenating or elementwise summing their feature representations. While computationally straightforward, this method often suffers from high-dimensional feature spaces and significant redundancy. Zhang et al. [39] developed a multisource feature selection module in the network to solve the problem of balancing HSI and LiDAR features in fusion and to reduce the generation of redundant information by dynamically filtering the source components in the sequence features. While feature selection-based fusion offers intuitive and computationally efficient results, this approach inevitably discards potentially discriminative information, ultimately compromising classification performance. A two-stage fusion strategy is used for feature fusion in the method proposed by Guo et al. [40]. The fusion framework initially employs sequence-to-sequence integration through multihead cross attention, subsequently incorporating low-frequency residuals to enhance intraclass feature representation and improve fusion robustness. Although effective, this approach suffers from

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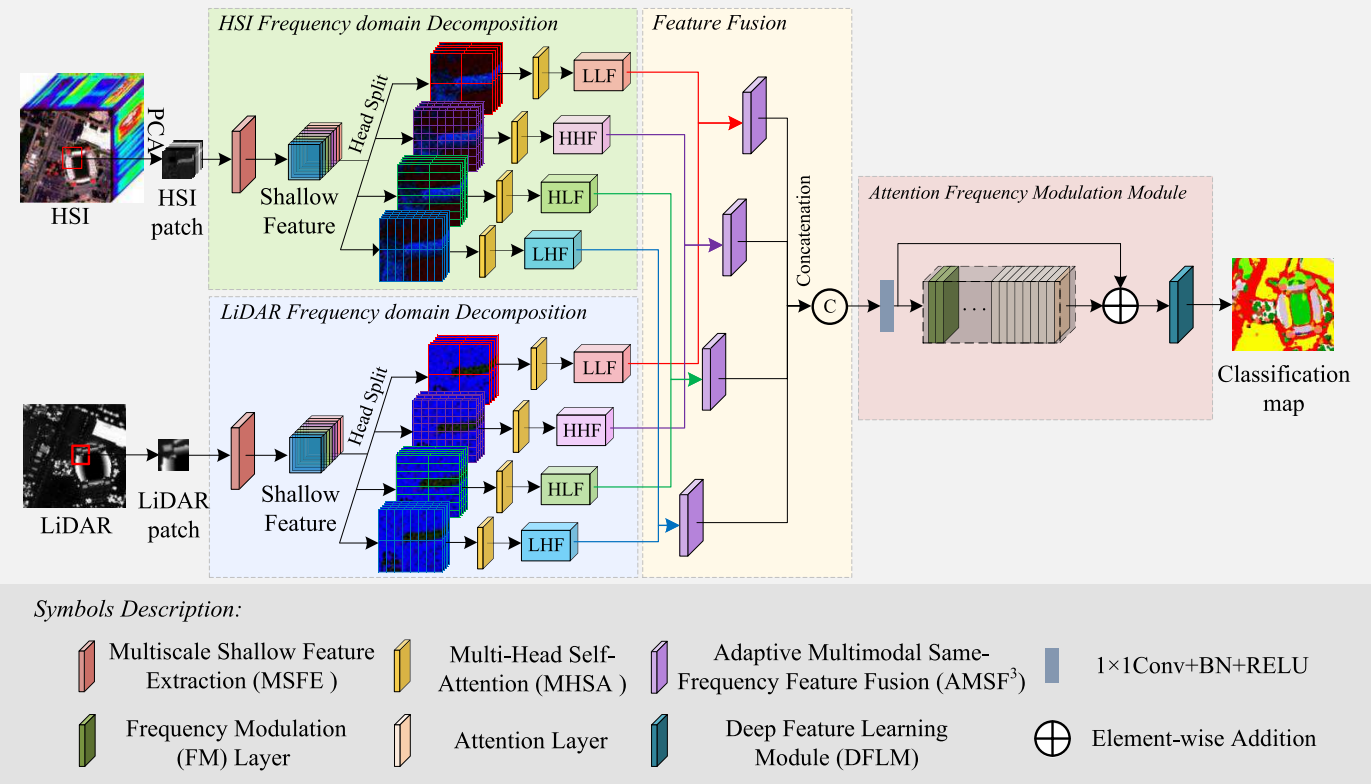


Fig. 2. Framework of the proposed method. MSFE is the multiscale shallow feature extraction module, AMSF^3 is the adaptive same-frequency feature fusion module, FM is the frequency modulation layer, and DFLM is the deep feature learning module.

unequal modal contributions to classification performance and demonstrates limited adaptability across varying application scenarios. We expect to enhance model performance by selectively emphasizing the most discriminative features during the fusion process. The AMSF^3 method accomplishes this through dynamic weighting of multimodal features coupled with autonomous learning of cross-modal relationships, ultimately improving both classification accuracy and generalization performance.

III. Proposed Network

A. Overall Architecture of the Proposed Network

The proposed frequency-domain transformer network (AF^{2}DN) is designed for joint classification of HSI and LiDAR data. By effectively leveraging complementary multimodal information, this approach overcomes the performance limitations inherent to single-modality data analysis.

For the AF^{2}DN networks, the framework is shown in Fig. 2. Specifically, the original HSI data $X_{\mathrm{HSI}} \in \mathbb{R}^{B\times H\times W}$ and the LiDAR data $X_{\mathrm{LiDAR}} \in \mathbb{R}^{1\times H\times W}$ were acquired by different sensors from the same area, where H and W represent the height and width of both types of data and B represents the spectral amount of HSI. As the hyperspectral data has some data redundancy, the PCA dimensionality reduction method is used in this article to obtain the dimensionality reduced hyperspectral data $X'_{\mathrm{HSI}} \in \mathbb{R}^{b\times H\times W}$, and b represents the dimensionality of the reduced dimensionality ($b \le B$). In order to improve the training efficiency, we performed patch division of HSI and LiDAR data, and after division, we obtained HSI data $X^p_{\mathrm{HSI}} \in \mathbb{R}^{b\times p\times p}$ and LiDAR data $X^p_{\mathrm{LiDAR}} \in \mathbb{R}^{1\times p\times p}$, where p represents the size of the patch. The rest of the AF^{2}DN consists of four main components, which are: multiscale shallow feature extraction (MSFE), TFFD, AMSF^3, and AFMM, which are described in detail in Sections III-B–III-E. Through this series of operations, AF^{2}DN can effectively fuse the advantageous information of multimodal data to achieve more accurate classification of remote sensing data.

B. Multiscale Shallow Feature Extraction

In multimodal data fusion classification, addressing scale variations across different data modalities while maintaining feature richness remains a critical challenge. To overcome this, we propose an MSFE module that employs multiscale convolutional operations and channelwise fusion to effectively capture fundamental features from each remote sensing data source.

The MSFE is shown in Fig. 3, taking HSI as an example. First, we input the hyperspectral patch X^p_{HSI} into three convolution modules with different scales, and the calculation is as follows:

$$M^i_{\mathrm{HSI}} = \mathrm{ReLU}\left(f^i_{\mathrm{BN}}\left(\mathrm{Con}^i_{s\times s}\left(X^p_{\mathrm{HSI}}\right)\right)\right) \tag{1}$$

where M^i_{HSI} represents the output feature of HSI at the ith scale (i = 1, 2, 3), $f^i_{\mathrm{BN}}(\cdot)$ represents the batch normalization function, and s represents the convolution kernel size, which we set to 3, 5, and 7, respectively.

Due to the spectral similarity between adjacent channels in HSI data, the convolutional operation preserves both

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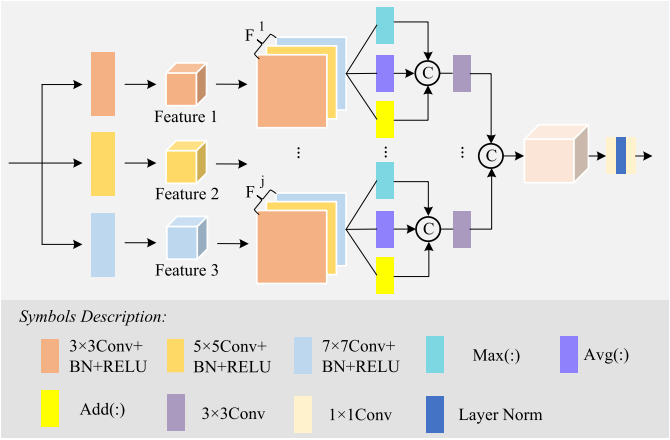


Fig. 3. Illustration of MSFE.

the channel order and spectral reflectance characteristics. Consequently, we perform channelwise concatenation of the multiscale features to generate the jth group's fused feature F_c^j representation

$$F_c^j = \text{Concat}\left(M_{\text{HSI}}^1(j), M_{\text{HSI}}^2(j), M_{\text{HSI}}^3(j)\right) \tag{2}$$

where j represents the jth channel. We further process F_c^j by performing elementwise summing, elementwise averaging, and max-pooling operations along the channel to extract channel-level attributes, generating three feature maps F_{add}^j, F_{avg}^j, and F_{max}^j, which are calculated as follows:

$$F_{\text{add}}^j = \text{Add}\left(F_c^j\right) \tag{3}$$

$$F_{\text{avg}}^j = \text{Avg}\left(F_c^j\right) \tag{4}$$

$$F_{\text{max}}^j = \text{Max}\left(F_c^j\right). \tag{5}$$

Next, to reduce the computational complexity, we use 3×3 convolution to obtain low-dimensional features F^j, calculated as follows:

$$F^j = \text{Conv}_{3\times3}\left(\text{Cat}\left(F_{\text{add}}^j, F_{\text{avg}}^j, F_{\text{max}}^j\right)\right). \tag{6}$$

Then, we concatenate the k groups of feature maps, perform layer normalization and 1×1 convolution on the fused feature maps, and obtain the final multiscale shallow feature F_H^S. The formula is as follows:

$$F_H^S = \text{Conv}_{1\times1}\left(f_{LN}\left(\text{Conv}_{1\times1}\left(\text{Concat}_{j\in\{1,2,\ldots,k\}}\left(F^j\right)\right)\right)\right). \tag{7}$$

Through the above operations, we obtain multiscale shallow features F_H^S and F_L^S, which provide rich basic features for subsequent feature learning.

C. Transformer-Based Frequency Feature Decomposition

Building upon the Transformer architecture, we propose a novel TFFD module. Current multimodal remote sensing approaches frequently neglect frequency-aware feature learning in their attention mechanisms. The standard self-attention operation inherently functions as a frequency-adaptive filter: smaller attention windows predominantly capture high-frequency details, while larger windows emphasize low-frequency components. To overcome the directional constraints of conventional frequency decomposition, we employ variably shaped attention windows to extract orientation-specific frequency components from hyperspectral and LiDAR data, enabling more efficient feature representation.

Specifically, first, for a given input shallow feature $F_H^S \in \mathbb{R}^{C\times p\times p}$ and $F_L^S \in \mathbb{R}^{C\times p\times p}$, we linearly project C channels into K heads in the channel dimension (here, we need to set K to an integer multiple of 4). We then divide the K heads in parallel into four groups of K/4 heads each, and will compute one type of self-attention per group. Next, we divide the shallow features F_H^S and F_L^S uniformly into four nonoverlapping windows, where the window corresponding to the low-frequency feature (LLF) contains $4s \times 4s$ tokens, the window corresponding to the high-frequency feature (HHF) contains $s \times s$ tokens, the window corresponding to the vertical features (HLF) contains $s \times 4s$ tokens, and the window corresponding to the horizontal features (LHF) contains $4s \times s$ tokens, and s is the window coefficient, a positive integer. Here, we take the HLF of HSI as an example, and the calculation process is as follows.

The HSI shallow features F_H^S are divided into M nonoverlapping windows $\lceil X^1, \ldots, X^M \rceil$, with the following formula:

$$F_H^S = \left(X^1, \ldots, X^M\right) \tag{8}$$

where $M = \dfrac{p \times p}{s \times 4s}$, M is the number of windows and each window contains $s \times 4s$ tokens.

Assuming that the queries, keys, and values of the kth $((1/2)K \le k \le (3/4)K)$ head are d_k-dimensional tensor $(d_k = (C/K))$, the attention output of the kth head can be obtained as follows:

$$Y_k^i = \text{Attention}\left(X^i W_k^Q, X^i W_k^K, X^i W_k^V\right) \tag{9}$$

$$\text{HLF}_k = \left(Y_k^1, Y_k^2, \ldots, Y_k^M\right) \tag{10}$$

where $X^i \in \mathbb{R}^{C\times s\times 4s}$, $i = 1, \ldots, M$, $W_k^Q, W_k^K, W_k^V \in \mathbb{R}^{C\times d_k}$ are the queries, keys, and values matrices of the kth header, respectively. Finally, the final HLF can be obtained by dimensional concatenating

$$\text{HLF}_{\text{HSI}} = \text{Concat}\left(\text{HLF}_{\frac{1}{2}K+1}, \text{HLF}_{\frac{1}{2}K+2}, \ldots, \text{HLF}_{\frac{3}{4}K}\right). \tag{11}$$

The other three types of attention for HSI and the four types of attention for LiDAR can be derived with reference to this process, and their outputs are LLF_{HSI}, HHF_{HSI}, LHF_{HSI}, $\text{LLF}_{\text{LiDAR}}$, $\text{HHF}_{\text{LiDAR}}$, $\text{HLF}_{\text{LiDAR}}$, and $\text{LHF}_{\text{LiDAR}}$, respectively. In this way, we have achieved the frequency feature decomposition of different modal data under the Transformer model.

D. Adaptive Multimodal Same-Frequency Feature Fusion

Following multifrequency feature extraction from HSI and LiDAR data, effective cross-modal fusion becomes crucial for successful classification. The effective feature fusion enhances cross-modal consistency while reducing intermodal discrepancies, addressing key challenges in high-dimensional remote sensing data, such as incomplete discriminative feature extraction and suboptimal fusion. This leads to significant improvements in classification performance.

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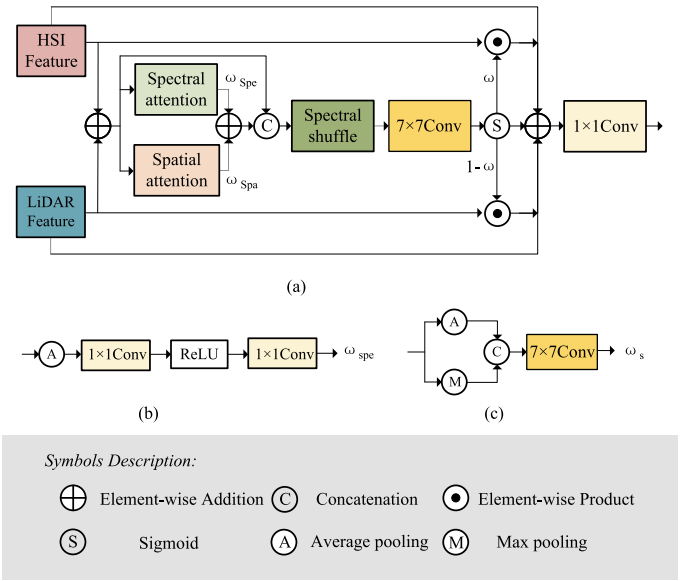


Fig. 4. AMSF^3 illustration. (a) AMSF^3 module framework. (b) Spectral attention. (c) Spatial attention.

A common feature fusion method is to fuse different modal features directly and linearly, such as element summing or direct concatenating, which results in a large information load and limits the feature expression capability of multimodal data. In this regard, we propose an AMSF^3 strategy, as shown in Fig. 4. This is a coarse-then-fine weight assignment fusion scheme that adaptively fuses the low-level features of the encoder part with the corresponding high-level features by learning the spatial weights to modulate the features. It is worth mentioning that the fusion method of same-frequency features can capture important feature information in the same-frequency range of each modal data, and this method can avoid the loss of sensitivity to some important features during data fusion, which in turn improves the ability of the classification model in recognizing complex features or detail information. Here, we take LLFs as an example, specifically, first, for the given HSI and LiDAR LLF features, $\mathrm{LLF}_{\mathrm{HSI}} \in \mathbb{R}^{C\times p\times p}$ and $\mathrm{LLF}_{\mathrm{LiDAR}} \in \mathbb{R}^{C\times p\times p}$, we perform an element summing operation to obtain the input features

$$X = \mathrm{LLF}_{\mathrm{HSI}} + \mathrm{LLF}_{\mathrm{LiDAR}}. \tag{12}$$

Then, we perform global average pooling in the channel dimension, global average pooling, and global maximum pooling in the spatial dimension, and go through the spectral attention module and the spatial attention module to obtain the spectral attention and spatial attention with the expressions

$$\omega_{\mathrm{spe}} = \mathrm{Conv}_{1\times1}\left(\max\left(0, \left(\mathrm{Conv}_{1\times1}\left(X^{C}_{\mathrm{GAP}}\right)\right)\right)\right) \tag{13}$$

$$\omega_{\mathrm{spa}} = \mathrm{Conv}_{7\times7}\left(\mathrm{Concat}\left(X^{S}_{\mathrm{GAP}}, X^{S}_{\mathrm{GMP}}\right)\right) \tag{14}$$

where ω_{spe} and ω_{spa} are the spectral weights and spatial weights, respectively, $\mathrm{Conv}_{1\times1}(\cdot)$ is a 1×1 convolutional layer, $\mathrm{Conv}_{7\times7}(\cdot)$ is a 7×7 convolutional layer, $\max(\cdot)$ is taken as the maximum value, and X^{C}_{GAP} represents the global average pooling of the input X in the channel dimension. X^{S}_{GAP} and X^{S}_{GMP} represent the input X for global average pooling and global maximum pooling in the spatial dimension, respectively.

Through an additive operation, we fused the spectral and spatial weights to obtain the coarse weights ω_{coa}

$$\omega_{\mathrm{coa}} = \omega_{\mathrm{spe}} + \omega_{\mathrm{spa}}. \tag{15}$$

Next, we rearrange the coarse weights ω_{coa} and each channel of the input X by a rearrangement operation to obtain the fine weights ω with the expression

$$\omega = \sigma\left(\mathrm{Gconv}_{7\times7}\left(\mathrm{CS}\left([X, \omega_{\mathrm{coa}}]\right)\right)\right) \tag{16}$$

where $\sigma(\cdot)$ is the sigmoid function, $\mathrm{Gconv}_{7\times7}(\cdot)$ is the group convolution, the number of groups is set to the number of channels, and $\mathrm{CS}(\cdot)$ is the channel rearrangement operation.

Finally, the same-frequency fusion feature LLFF under the LLF is obtained using fine weights, combined with weighted summation and residual connection

$$\mathrm{LLFF} = \mathrm{Conv}_{1\times1}\left(X + \omega\mathrm{LLF}_{\mathrm{HSI}} + (1-\omega)\,\mathrm{LLF}_{\mathrm{LiDAR}}\right). \tag{17}$$

Multiple same-frequency fusion features are concatenated and fused to obtain the final multimodal fusion feature F_{fuse}

$$F_{\mathrm{fuse}} = \mathrm{Concat}\left(\mathrm{LLFF}, \mathrm{HHFF}, \mathrm{HLFF}, \mathrm{LHFF}\right). \tag{18}$$

The AMSF^3 module adaptively fuses cross-modal frequency features, enhancing the model's capacity to extract and integrate discriminative information from heterogeneous data sources. This results in significantly improved classification accuracy and robustness.

E. AFM Module

Following frequency-domain decomposition and AMSF^3 integration, the resulting features comprehensively incorporate both low- and high-frequency components. Frequency components exhibit distinct contributions to multimodal fusion classification. LLFs primarily characterize large-scale patterns, including macroscopic structures, global shapes, and surface typologies, as they encode smooth, wide-range variations. Conversely, high-frequency components preserve fine-grained details, such as textural patterns and edge information. To enable the selective frequency component weighting and minimize intercomponent redundancy, we propose AFM module (AFMM). The proposed framework integrates FM with attention mechanisms to enhance the feature learning. The block-based FM layer dynamically modulates frequency components via a learnable weight matrix to enhance the local feature extraction, while the attention layer selectively processes global contextual and semantic information.

For the FM layer, its structure is shown in Fig. 5. First, we apply a block-based fast Fourier transform (FFT) to transform the features F_{ffn} from the spatial domain to the frequency domain. Then, the weight learning matrix W is introduced to modulate different frequency components to obtain the FM features F_{fm}

$$F_{\mathrm{ffn}} = \mathrm{Conv}_{1\times1}\left(\left(\mathrm{GELU}\left(\mathrm{Conv}_{1\times1}\left(f_{\mathrm{LN}}\left(F_{\mathrm{in1}}\right)\right)\right)\right)\right) \tag{19}$$

$$F_{\mathrm{fm}} = \Gamma\left(\mathrm{B}\left(F_{\mathrm{ffn}}\right)\right) \odot W \tag{20}$$

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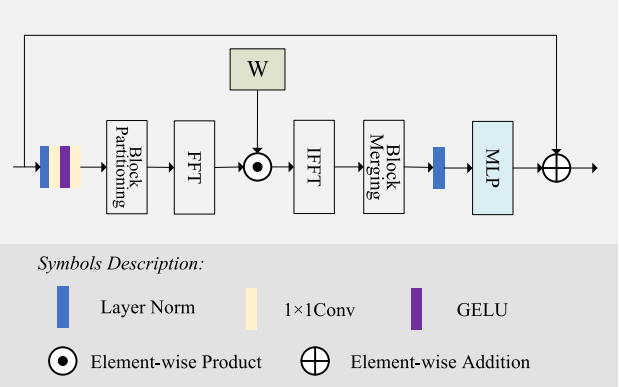


Fig. 5. Illustration of the FM layer.

where F_{in1} is the input features of FM layer, $f_{\text{LN}}(\cdot)$ is the layer normalization, W represents the weight learning matrix, $\odot$ represents the elemental multiplication, $B(\cdot)$ represents the block partitioning operation, and $\Gamma(\cdot)$ represents FFT.

Subsequently, the inverse FFT (IFFT) was used on F_{fm} and rearranged to obtain the refined features F_{out1}

$$F_{\text{out1}} = \text{MLP}\left(\text{B}^{-1}\left(\Gamma^{-1}\left(F_{\text{fm}}\right)\right)\right) \tag{21}$$

where F_{out1} is the output feature of the FM layer, $B^{-1}(\cdot)$ represents the block merging operation, $\Gamma^{-1}(\cdot)$ denotes the IFFT, and MLP$(\cdot)$ is the multilayer perceptron operation.

The structure of the attention layer is relatively simple, and only requires combining the traditional multihead self-attention, MLP, and residual connection methods, as follows:

$$\begin{aligned} F_{\text{out2}} = {} & F_{\text{in2}} + \text{MHSA}\left(f_{\text{LN}}\left(F_{\text{in2}}\right)\right) \\ & + \text{MLP}\left(f_{\text{LN}}\left(F_{\text{in2}} + \text{MHSA}\left(f_{\text{LN}}\left(F_{\text{in2}}\right)\right)\right)\right) \end{aligned} \tag{22}$$

where F_{out2} is the output feature of the attention layer, MHSA$(\cdot)$ is the multihead self-attention operation, and $f_{\text{LN}}(\cdot)$ is the layer normalized.

The AFM module we proposed is to pass the multimodal fusion feature F_{fuse} through the superimposed FM layer and attention layer in sequence to achieve the purpose of local and global feature learning. The proportion of FM layers in the total number of layers can be controlled by setting the α factor, when α is higher it means that the model pays more attention to local details in the data, which means that it is favorable for tasks that require accurate identification of details and processing of fine-grained information, and α is lower it means that the model pays more attention to the overall structure in the data, which is effective for capturing global information and understanding large-scale changes. By adjusting α appropriately, the model is able to achieve a more refined classification.

In order to further highlight the key important information, we pass the fusion feature F through a deep feature learning module (DFLM). The detailed process is as follows:

$$F_{\text{deep}} = \text{Conv}_{1\times1}(F) \otimes \sigma\left(\text{Conv}_{1\times1}\left(\text{Conv}_{1\times1}(F)\right)\right) \tag{23}$$

where $\sigma(\cdot)$ represents the sigmoid activation function and $\otimes$ represents the tensor multiplication operation.

Finally, with the fully connected and softmax functions, we can convert the fused features into probability distributions, where the labels with the highest probability represent the land cover classes.

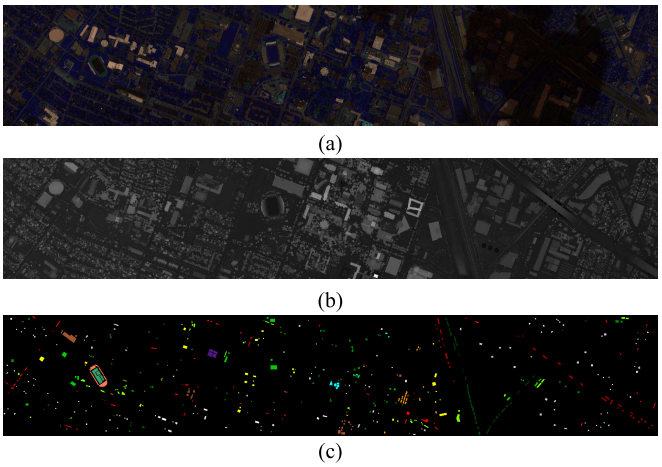


Fig. 6. Houston2013 dataset. (a) HSI pseudo-color image representation (bands 30, 70, and 100). (b) DSM-based LiDAR image representation. (c) Ground truth.

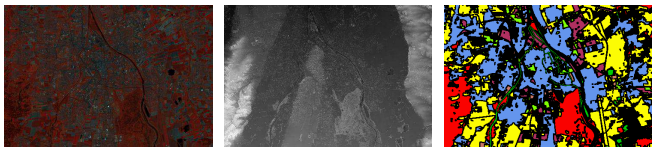


Fig. 7. Augsburg dataset. (a) HSI pseudo-color image representation (bands 50, 100, and 150). (b) DSM-based LiDAR image representation. (c) Ground truth.

IV. Experiments

In this section, we validate the performance of the proposed AF²DN method by conducting experiments on three widely used benchmark datasets: Houston2013, Augsburg, and GRSS-DFC-2007. First, we briefly describe these datasets and the experimental setup. Then, the parameters of AF²DN are analyzed in detail. Subsequently, we compare AF²DN with other state-of-the-art methods to assess its superiority. Finally, the effectiveness of each component of the method was evaluated through ablation experiments.

A. Datasets Description

1) Houston2013: The Houston2013 dataset was acquired by the National Center for Airborne Laser Mapping on the University of Houston campus and was designed to acquire HSI- and digital surface model (DSM)-based LiDAR data. Both HSI and LiDAR data have a spatial resolution of 2.5 m and contain a total of 349 × 1905 pixels, with the HSI data containing 144 bands ranging from 0.38 to 1.05 μm. The dataset is labeled with 15 029 ground-truth samples covering samples selected from 15 categories. Fig. 6 shows the HSI pseudo-color images (bands 30, 70, and 110), the DSM-based LiDAR representation and the ground truth under the Houston2013 dataset. The division details of the training and testing sets are given in Table I.

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

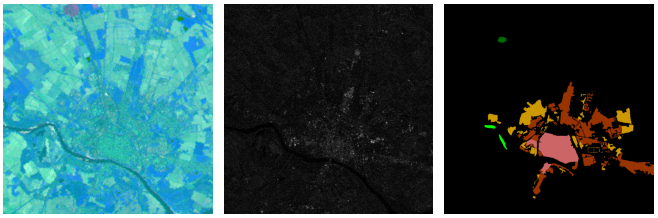
No.	Class(Train/Test)	CoupledCNNs	LSAF	ExViT	CALC	HCT	NCGLF^2	FDNet	AM^{3}Net	AF^{2}DN
1	Health grass(20/1231)	96.86	90.51	**97.97**	97.88	92.36	95.51	98.78	97.89	97.47
2	Stressed grass(20/1234)	97.43	97.91	94.49	98.7	92.14	96.09	96.52	94.33	**98.84**
3	Synthetic grass(20/677)	97.59	**99.79**	95.86	97.34	98.97	96.18	96.31	98.38	98.50
4	Trees(20/1224)	99.41	81.29	96.24	91.33	99.67	96.54	93.95	**99.75**	98.21
5	Soil(20/1222)	99.05	99.38	99.74	99.82	99.10	99.33	99.59	99.75	**99.83**
6	Water(20/305)	98.65	99.60	97.70	**100.00**	99.67	97.30	96.06	96.39	98.26
7	Residential(20/1248)	96.19	70.74	94.23	95.27	90.54	89.73	96.07	87.34	**96.50**
8	Commercial(20/1224)	91.51	85.20	87.01	**94.44**	87.75	84.13	78.76	85.21	91.88
9	Road(20/1232)	89.40	84.90	83.93	89.85	83.93	**91.31**	80.68	87.66	88.91
10	High way(20/1207)	95.30	97.17	89.73	95.60	80.86	99.47	97.35	92.29	**99.57**
11	Rail way(20/1215)	96.04	96.74	91.44	95.39	96.87	98.26	88.06	95.88	**98.34**
12	Parking_lot_1(20/1213)	88.24	90.14	**98.02**	95.54	77.66	96.90	96.04	94.39	97.94
13	Parking_lot_2(20/449)	95.28	**96.08**	95.10	95.54	93.10	91.10	94.43	91.76	95.05
14	Tennis court(20/408)	**100.00**	99.85	**100.00**	**100.00**	99.38	**100.00**	**100.00**	98.53	99.97
15	Running track(20/640)	99.89	**100.00**	**100.00**	99.68	**100.00**	**100.00**	99.84	98.44	**100.00**
	OA(%)	95.52	91.01	94.03	95.89	92.63	95.14	93.40	94.03	**96.83**
	AA(%)	96.05	92.62	94.77	95.55	94.04	95.49	94.16	94.53	**97.09**
	Kappa×100	95.15	90.27	93.55	96.43	92.04	94.74	92.86	93.54	**96.59**

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE AUGSBURG DATASET

No.	Class(Train/Test)	CoupledCNNs	LSAF	ExViT	CALC	HCT	NCGLF^2	FDNet	AM^{3}Net	AF^{2}DN
1	Forest(344/13163)	98.71	99.06	96.23	95.94	98.11	96.92	**99.19**	98.53	98.78
2	Residential-Area(334/29995)	97.04	94.81	95.87	96.55	94.10	95.81	94.02	95.49	**97.30**
3	Industrial-Area(112/3739)	86.95	85.21	74.11	87.35	85.45	83.83	84.48	83.77	**87.68**
4	Low-Plants(1162/25695)	98.87	97.25	97.98	97.71	99.05	98.66	98.63	**99.27**	98.86
5	Allotment(91/484)	94.21	95.87	54.75	76.24	93.18	94.65	93.38	95.66	**95.94**
6	Commercial-Area(23/1622)	3.7	27.74	27.44	0.00	29.28	**39.30**	16.27	6.97	37.35
7	Water(312/1218)	90.23	89.43	83.25	83.83	89.82	88.79	90.11	89.08	**90.25**
	OA(%)	95.38	95.05	93.65	94.75	94.58	95.05	94.32	94.72	**96.21**
	AA(%)	81.42	83.91	75.66	80.16	84.14	85.42	82.60	81.25	**86.59**
	Kappa×100	93.38	92.92	90.88	92.48	92.28	92.90	91.92	92.44	**94.57**

2) Augsburg: The Augsburg dataset was collected from the city of Augsburg, Germany, and the data sources include satellite-based HSI and DSM-based LiDAR data. The HSI and LiDAR data were provided by DAS-EOC and DLR-3K. The dataset has a scene spatial resolution of 30 m and consists of a total of 332 × 485 pixels. The HSI data consists of 180 bands ranging from 0.4 to 2.5 μm, covering a wide range of wavelength bands from the visible to near infrared. The ground reference data are provided by OpenStreetMap to ensure high accuracy and reliability. This dataset is widely used in remote sensing data processing, feature classification, and urban and terrain analysis. Fig. 7 shows the HSI pseudo-color images (bands 50, 100, and 150), the DSM-based LiDAR representation and the ground truth under the Augsburg dataset. The detailed data division in the experiment is shown in Table II.

3) GRSS-DFC-2007: The GRSS-DFC-2007 dataset consists of multispectral imagery (MSI) and synthetic aperture radar (SAR) images. The MSI data were acquired on April 7, 1994, with the Landsat-5 sensor, while the SAR images were acquired on October 3, 1994, with the ERS-1 sensor. The dataset is located in the Pavia region of Northern Italy and contains a total of 787 × 787 pixels and is labeled for five different classes of features. This dataset provides rich test data for remote sensing data fusion and feature classification tasks, which are widely used in remote sensing technology research and applications. Fig. 8 depicts the corresponding MSI and SAR images and the ground truth of the GRSS-DFC-2007



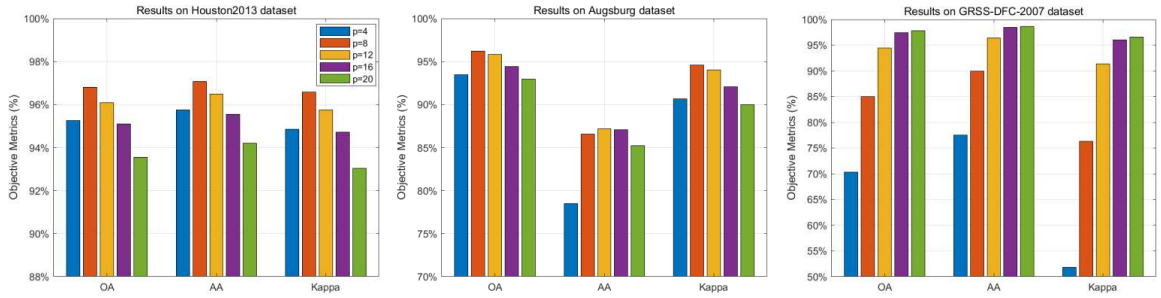
(a) (b) (c)

Fig. 8. GRSS-DFC-2007 dataset. (a) MSI pseudo-color image representation (bands 2, 4, and 6). (b) SAR data representation. (c) Ground truth.

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TABLE III
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON THE GRSS-DFC-2007 DATASET

No.	Class(Train/Test)	CoupledCNNs	LSAF	ExViT	CALC	HCT	NCGLF^2	FDNet	AM^{3}Net	AF^{2}DN
1	City Center(674/12811)	93.42	90.26	92.75	98.01	93.51	97.57	80.59	95.77	**98.95**
2	Residential-Area(1481/28136)	92.37	90.06	90.75	96.32	94.23	95.75	82.69	94.55	**97.64**
3	Sparse Building(752/14282)	84.60	77.74	79.33	91.54	91.58	88.28	63.25	88.5	**96.90**
4	Water(45/848)	95.94	97.44	99.76	100.00	96.46	99.90	95.99	90.45	**100.00**
5	Vegetation(30/561)	99.93	96.04	100.00	100.00	85.74	100.00	98.93	97.68	**100.00**
	OA(%)	90.77	87.17	88.55	95.59	93.35	94.38	77.68	93.27	**97.80**
	AA(%)	93.25	90.31	92.52	97.17	92.30	96.30	84.29	93.39	**98.69**
	Kappa×100	85.47	79.71	81.94	93.07	89.55	91.18	64.47	89.42	**96.57**



(a) (b) (c)

Fig. 9. Impact of different patch sizes on algorithm performance. (a) Houston2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

dataset. The exact segmentation of the training and testing data is shown in Table III.

B. Experimental Setting

1) Experimental Details: Our proposed AF^{2}DN method is implemented in the PyTorch framework, and performed on a PC server having a RAM of 32 GB, a CPU of i9-11900K and NVIDIA GTX3070Ti. To optimize the network, we use the Adam optimizer with a learning rate decay parameter of 0.9. The training batch size is set to 64, the number of epochs is set to 100, and the initial learning rate is set to 1e^{-4}. The window coefficient is set to 1.

2) Evaluation Metrics: In order to comprehensively evaluate the performance of AF^{2}DN classification methods, we adopted four widely used evaluation metrics: overall accuracy (OA), average accuracy (AA), kappa coefficient (Kappa), and classwise accuracy. To ensure the fairness and robustness of the experimental results, all experiments were conducted under the same conditions, and each evaluation was repeated and executed ten times, and its mean and standard deviation were calculated. The best performing method will be marked in bold in the text.

C. Parameter Analysis

1) Impact of Patch Size: In the multimodal remote sensing data fusion classification task, the patch size has a significant impact on the classification results. The choice of patch size involves a tradeoff between capturing sufficient context and preserving fine-grained details. Larger patches provide richer contextual and semantic information from surrounding pixels, which is crucial for distinguishing between semantically similar classes (e.g., "grass" versus "tree" that may have similar spectral signatures but different spatial arrangements). Conversely, smaller patches more effectively preserve high-frequency spatial details and sharp boundaries, enabling accurate segmentation of small objects and precise delineation of edges between classes. As shown in Fig. 9, we performed a parametric analysis of the effect of patch size on the model classification results under different datasets. Considering the effect of model frequency features, we take the variation of patch size from 4 to 20 with a growth interval of 4. As can be seen from the figure, the Houston2013 dataset and the Augsburg dataset show a trend of increasing and then decreasing, with the maximum peak occurring at 8×8. As for the GRSS-DFC-2007 dataset, there is a continuous increase, but the trend slows down significantly after 12×12, and we analyzed that this may be caused by the relative sparseness of land cover elements in the GRSS-DFC-2007 dataset. Based on the experimental results, we set the patch size to 8×8 for the Houston2013 dataset and Augsburg dataset and 20×20 for the GRSS-DFC-2007 dataset.

2) Impact of Data Dimension: In Fig. 10, we analyze the impact of data dimension on three different datasets. In fusion classification tasks, high-dimensional data such as HSI/MSI will increase the complexity of model training. At the same time, the model may face feature redundancy problems and even introduce noise, affecting the final classification performance. However, a dimension that is too low will result in the loss of key information and may not be sufficient to distinguish similar categories. As can be seen from the figure, when the dimension of the Houston2013 dataset is 10, and

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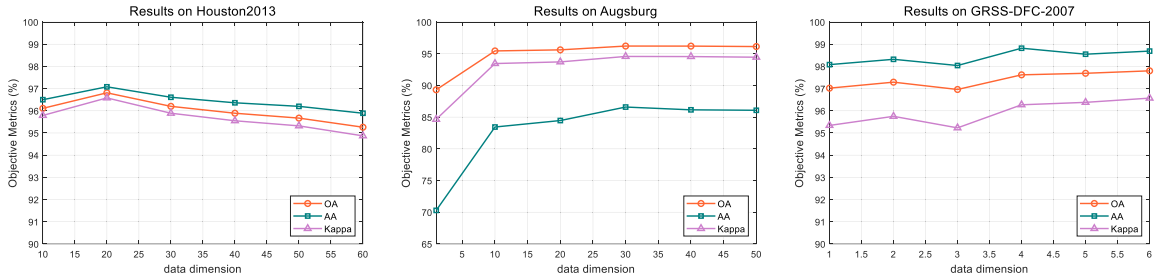


Fig. 10. Impact of different data dimensions on algorithm performance. (a) Houston2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

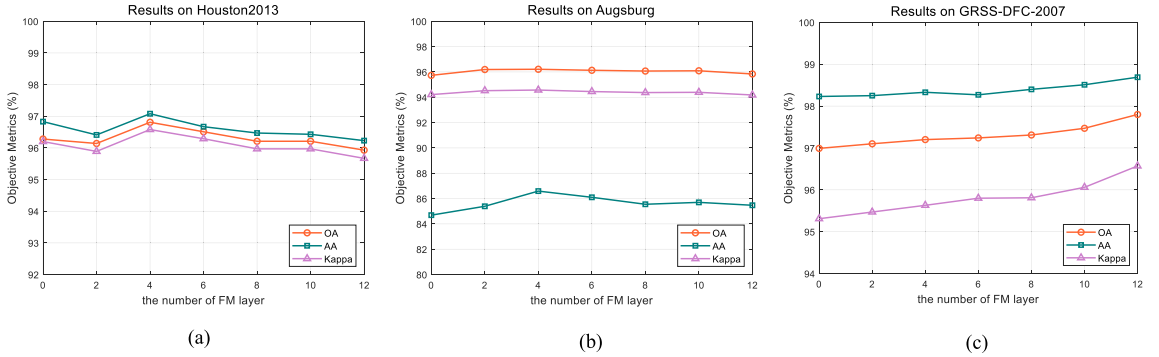


Fig. 11. Impact of different numbers of FM layers on algorithm performance. (a) Houston2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

the dimensions of the Augsburg dataset and the GRSS-DFC-2007 dataset are 1, significant information loss occurs. As the dimension gradually increases, in the Houston2013 dataset, the dimension peaks at 20 and then shows a downward trend. The peak in the Augsburg dataset is 30, and then, the classification performance tends to be stable. In the GRSS-DFC-2007 dataset, since MSI has only six bands and the fluctuation is not significant, we think it is reasonable to keep the dimension unchanged. Therefore, based on the experimental results, we set the dimension of the Houston2013 dataset to 20, the dimension of the Augsburg dataset to 30, and the dimension of the GRSS-DFC-2007 dataset to 6.

3) Impact of the Number of FM Layers: The number of FM layers in the feature learning layer, as part of the AFMM, also has a different impact on classification performance. A higher number of FM layers means that our network pays more attention to the high-frequency information and fine-grained properties of the image, while a lower number of FM layers makes the network more inclined to capture low-frequency information and global properties. As shown in Fig. 11, we illustrate the difference in classification performance using the number of FM layers in AF^{2}DN. We choose an architecture similar to SpectFormer with 12 feature learning layers and replace the attention layer with FM layers to achieve the purpose of changing the number of FM layers. It can be seen from the Houston2013 dataset and the Augsburg dataset that the peak of the FM layer appears at 4, and then slowly decreases and stabilizes as the number of layers increases. The classification performance index of the GRSS-DFC-2007 dataset gradually increases as the number of FM layers increases. According to the experimental results, we set the number of FM layers to 4 for the Houston2013 dataset and the Augsburg dataset, and the number of FM layers to 12 for the GRSS-DFC-2007 dataset.

D. Performance Comparison

In this section, first, we select some advanced multisource algorithms in recent years to perform performance comparisons and visualization results analysis on classification accuracy with our proposed method on three datasets. The selected algorithms include: CoupledCNNs, linear self-attention fusion (LSAF), extended Vision Transformer (ExViT), coupled adversarial learning-based classification (CALC), hierarchical CNN and Transformer (HCT), network combining global and local features for fusion of multisource remote sensing data (NCGLF^2), frequency-domain-based network (FDNet), adaptive mutual-learning-based multimodal data fusion network (AM^{3}Net) [41], [42], [43], [44], [45], [46], [47], [48]. These algorithms are benchmarks for comparison with our proposed AF^{2}DN method, and the parameters of all compared methods are set as described in their original papers. Then, for the fairness and effectiveness of the experiment, we randomly selected all training samples and test samples without overlapping. In addition, we performed a computational complexity analysis to evaluate the computational efficiency of various algorithms. At the end of this section, we also performed an analysis of the effect of training sample size on different datasets to demonstrate the advancedness of the proposed algorithm.

1) Results on Houston2013 Dataset: On the Houston2013 dataset, first, as shown in Table I, the classification results

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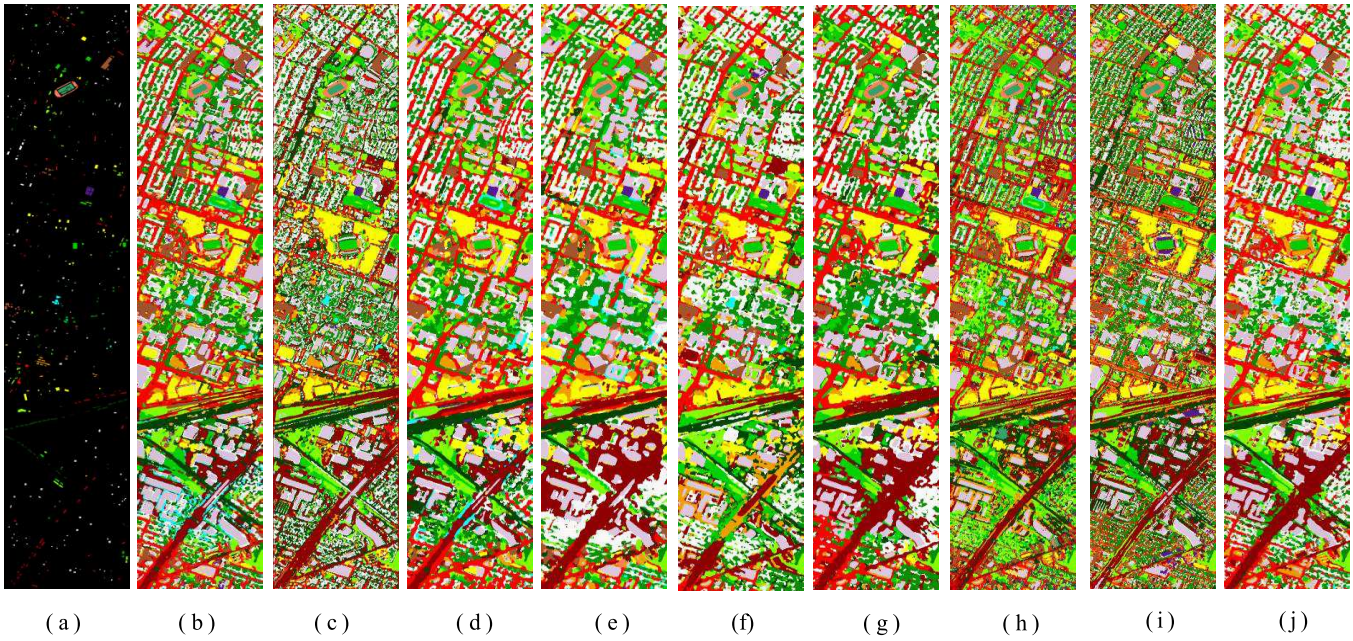


Fig. 12. Classification maps of different methods on the Houston2013 dataset. (a) GT. (b) Coupled CNNs. (c) LSAF. (d) ExViT. (e) CALC. (f) HCT. (g) NCGLF^2. (h) FDNet. (i) AM^{3}Net. (j) AF^{2}DN.

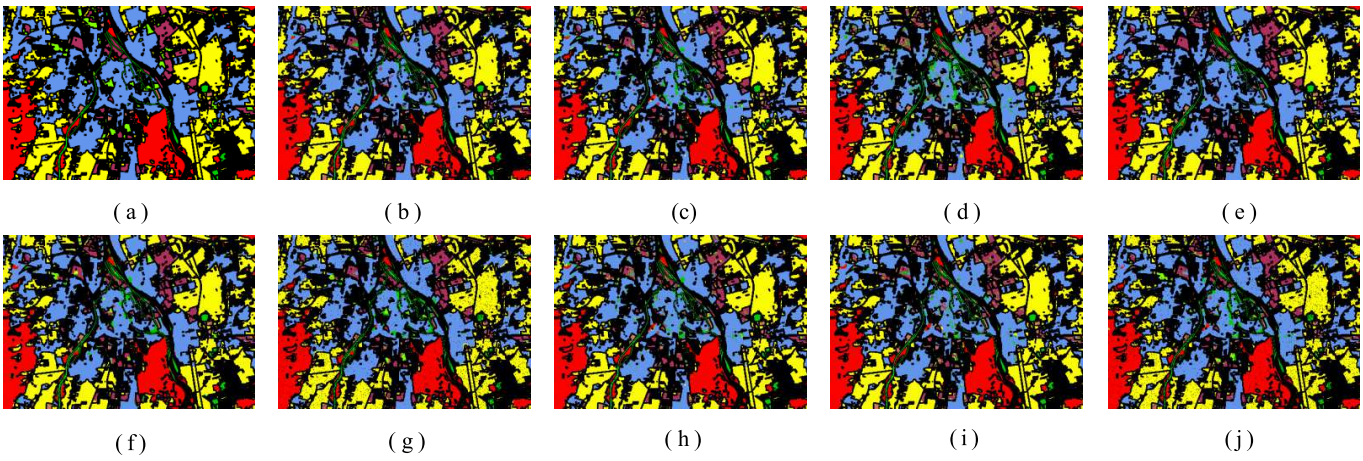


Fig. 13. Classification maps of different methods on the Augsburg dataset. (a) GT. (b) Coupled CNNs. (c) LSAF. (d) ExViT. (e) CALC. (f) HCT. (g) NCGLF^2. (h) FDNet. (i) AM^{3}Net. (j) AF^{2}DN.

of different methods for HSI and LiDAR data are shown. We use OA, AA, Kappa, and classwise accuracy to quantitatively describe the performance of classification. Then, as shown in Fig. 12, the visual classification maps of different methods are given. These classification maps can intuitively present the spatial distribution and category boundaries of the classification results, helping us quickly evaluate the model performance, identify the classification error area, and verify the rationality of the object category.

Houston2013 dataset has 15 land cover categories. Since some areas are affected by clouds and shadows, this puts high demands on the accuracy and robustness of the algorithm. First, overall, compared with all the comparison methods, our proposed AF^{2}DN has achieved the highest OA, AA, and Kappa, especially OA and AA, which are 0.94% and 1.04% higher than the comparison method with the highest accuracy. Second, our AF^{2}DN method has achieved the highest accuracy in 6 of the 15 land cover categories, and the accuracy in the Running track category has reached 100%. Although there are still nine categories that are not the most accurate, it performs comparable to the best category in most categories, such as health grass, synthetic grass, trees, Parking_lot_1, Parking_lot_2, and tennis court. Then, compared with CoupledCNNs, a deep CNN-based architecture, and HCT, a common CNN–Transformer architecture, the proposed method has a leading position in almost all category accuracy and OA indicators. From the classification diagram in Fig. 13, we can see that in areas blocked by clouds and shadows, the classification visualization results of the proposed AF^{2}DN method in the high way and rail way categories are smoother and closer to the labeled samples, and the category boundary of residential is relatively clearer, which further verifies the results in Table I. The above phenomena show that compared with other methods, the proposed AF^{2}DN has higher

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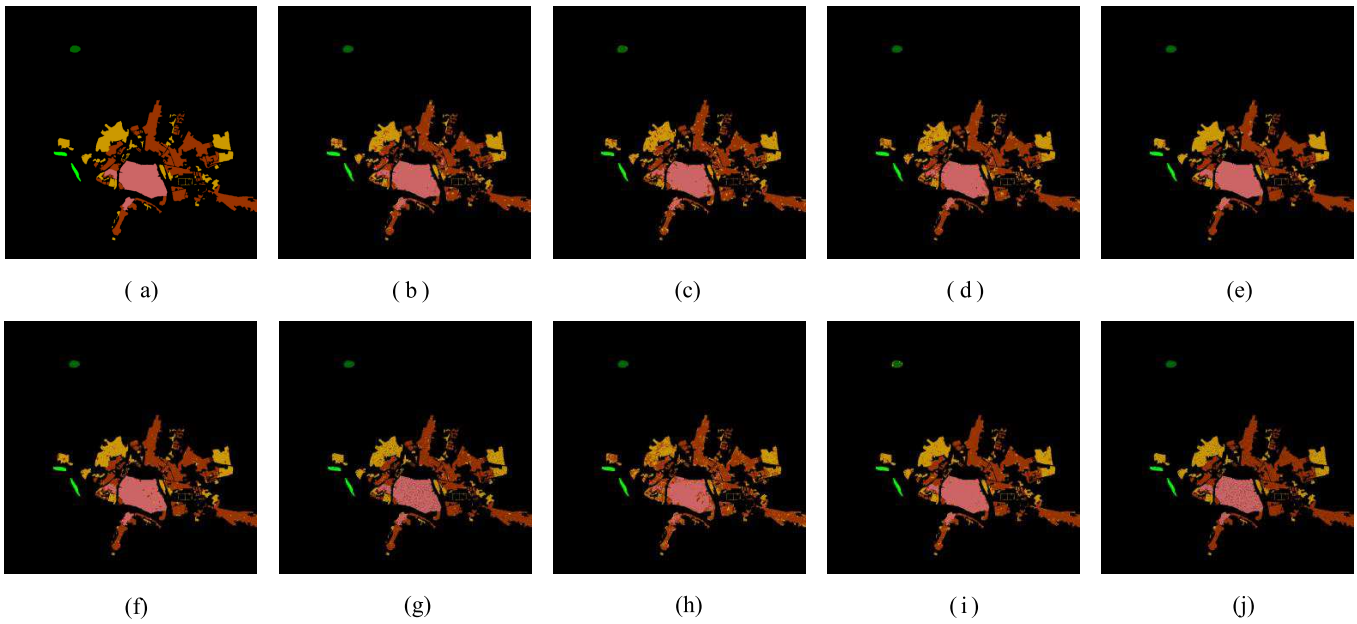


Fig. 14. Classification maps of different methods on the GRSS-DFC-2007 dataset. (a) GT. (b) Coupled CNNs. (c) LSAF. (d) ExViT. (e) CALC. (f) HCT. (g) NCGLF^2. (h) FDNet. (i) AM^{3}Net. (j) AF^{2}DN.

classification performance and better adaptability in complex environments covered by clouds and shadows.

2) Results on Augsburg Dataset: For the Augsburg dataset, which has seven different types of land cover categories, Table II shows the quantitative classification results, and Fig. 13 gives the classification maps of different methods. It can be seen that the AF^{2}DN we proposed has achieved the best results in the three indicators of OA, AA, and Kappa, leading the second highest method by 0.83%, 1.17%, and 1.19%, respectively, and has the highest accuracy in four categories. It is obvious that the classification accuracy of commercial area is less than 40% on almost all methods, which may be due to the limitation of the number and distribution of training samples. In addition, traditional Transformer-based methods such as ExViT and HCT lag behind our method in almost all indicators, which reflects the advantages of directional frequency feature extraction and FM. CALC is an unsupervised learning method that shows excellent indicators on large samples but inefficient performance on small samples in this dataset. It is worth noting that NCGLF^2 and FDNet, which also use frequency-domain modeling, perform relatively well and achieve the best classification results in forest and commercial area. Our proposed AF^{2}DN lags behind by only 0.41% and 1.95% in these two categories and shows better performance in other aspects. This fully illustrates the importance of the same-frequency adaptive fusion and FM to further eliminate data redundancy. Compared with the comparative methods, the AF^{2}DN algorithm demonstrates better effectiveness and robustness for imbalanced and dispersed samples.

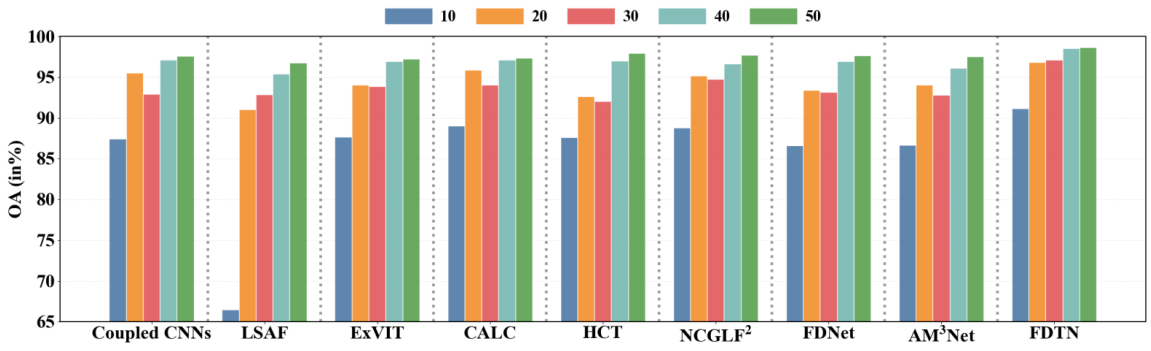
3) Results on GRSS-DFC-2007 Dataset: In order to expand the adaptability of multimodal data, we further validated our method using MSI and SAR on the GRSS-DFC-2007 dataset. GRSS-DFC-2007 focuses on the classification of objects in urban areas, testing the ability of data fusion to improve classification accuracy in complex scenes. Fig. 14 and Table III show the quantitative classification results and the visual classification map results, respectively. It can be seen that our AF^{2}DN method has achieved the highest results in all indicators, among which water and vegetation present perfect classification results. Compared with other algorithms, AF^{2}DN has a more accurate spatial distribution and relatively less noise, especially in the sparse building category. In general, satisfactory results were obtained on the GRSS-DFC-2007 dataset, verifying the potential of AF^{2}DN in fine urban classification.

4) Computational Complexity Analysis: Table IV lists the comparison of the computational complexity of the AF^{2}DN algorithm with various other algorithms on the Houston2013 dataset. To ensure a standardized and hardware-independent evaluation of computational efficiency, we employ two alternative metrics: the number of trainable parameters and the million floating-point operations (MFLOPs) required for the backpropagation stage. It can be observed that Coupled-CNN, a CNN-based algorithm, has fewer parameters and a lower MFLOPS count than the proposed AF^{2}DN. However, LSAF and AM^{3}Net, due to their use of interactive attention mechanisms for high-dimensional features across multiple stages, contain a relatively large number of trainable parameters. Compared with pure Transformer-based methods (such as ExViT), algorithms that integrate Transformers with frequency-domain concepts—such as NCGLF^2, FDNet, and AF^{2}DN—demonstrate greater computational efficiency. Furthermore, CNN–Transformer hybrid models typically possess more trainable parameters than CNN-based architectures, which is attributable to the inclusion of the Transformer's attention mechanism. Notably, the proposed AF^{2}DN exhibits lower algorithmic complexity than LSAF and NCGLF^2, which also employ attention and Transformer structures. Although its floating-point operation count is slightly higher than that

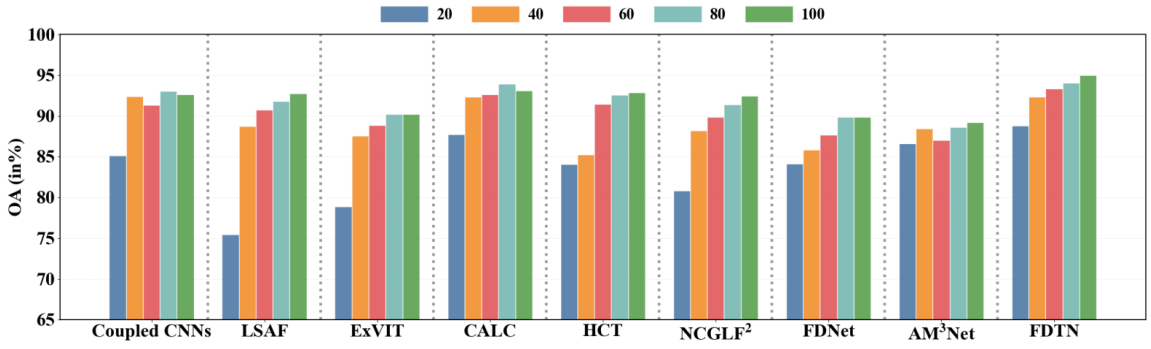
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TABLE IV
COMPARISON OF COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

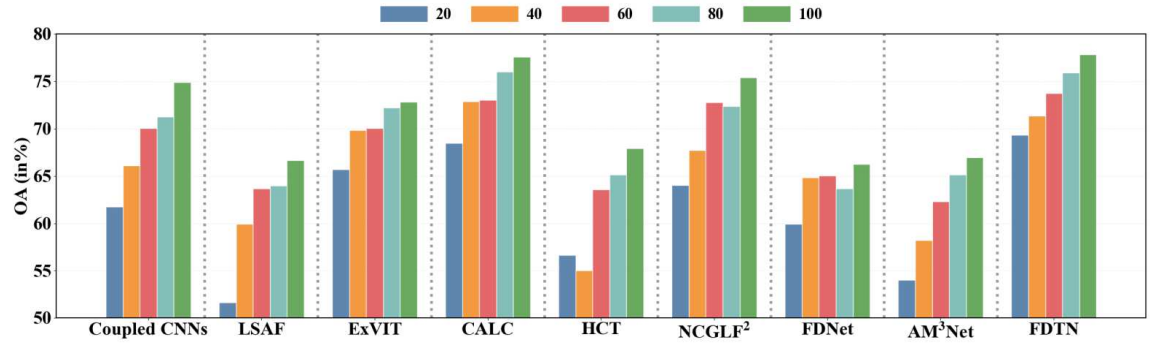
Methods	CoupledCNNs	LSAF	ExViT	CALC	HCT	NCGLF^2	FDNet	AM^{3}Net	AF^{2}DN
Params(M)	0.105	2.157	0.290	0.284	0.430	0.390	0.346	1.057	0.351
Computation Cost(MFLOPs)	2.96	105.49	4189.35	37.05	14.87	116.48	15.05	10.15	35.91



(a)



(b)



(c)

Fig. 15. Comparison of classification performance of various methods on three datasets with different numbers of training samples. Different numbers of training samples are represented by different colors. (a) Houston2013 dataset. (b) Augsburg dataset. (c) GRSS-DFC-2007 dataset.

of HCT, FDNet, and AM^{3}Net, we consider this tradeoff acceptable given the resultant improvement in performance. Experimental results demonstrate that AF^{2}DN achieves an optimal balance between computational efficiency and classification accuracy in HSI and LiDAR data classification tasks, demonstrating good practical application value and broad development prospects.

5) Analysis of the Role of Training Sample Size: The stability brought by the change of training samples is also a problem that often receives attention. In order to further verify the classification performance of our AF^{2}DN with different training numbers, Fig. 15 shows the performance comparison of various methods on three datasets with the change of training sample numbers. Specifically, on the Houston2013

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TABLE V
ABLATION EXPERIMENTS OF DIFFERENT MODULES ON THREE DATASETS

MSFE	LLF	HHF	HLF	LHF	AMSF^3	AFM	Houston 2013			Augsburg			GRSS-DFC-2007		
							OA(%)	AA(%)	Kappa×100	OA(%)	AA(%)	Kappa×100	OA(%)	AA(%)	Kappa×100
×	✓	✓	✓	✓	✓	✓	96.01	96.57	95.68	95.83	82.00	93.99	97.31	97.97	96.35
✓	×	✓	✓	✓	✓	✓	95.38	95.46	93.92	95.76	84.85	93.93	97.53	98.02	96.49
✓	✓	×	✓	✓	✓	✓	96.81	97.08	96.55	96.18	86.56	94.55	97.76	98.67	96.56
✓	✓	✓	×	✓	✓	✓	95.73	96.31	95.38	95.87	85.38	94.09	97.57	98.09	96.46
✓	✓	✓	✓	×	✓	✓	95.43	95.86	95.06	95.81	82.90	94.01	97.56	98.15	96.54
✓	✓	✓	✓	✓	×	✓	94.60	95.58	94.16	95.23	85.72	93.60	96.92	97.77	95.64
✓	✓	✓	✓	✓	✓	×	95.83	96.50	95.49	95.87	86.25	94.08	97.34	98.13	96.01
✓	✓	✓	✓	✓	✓	✓	96.83	97.09	96.59	96.21	86.59	94.57	97.80	98.69	96.57

dataset, we took the training number interval of 10 from 10 to 50 and conducted five sets of experiments. For the Augsburg dataset and GRSS-DFC-2007 dataset, we took the training number interval of 20 from 20 to 100.

From the results, we can see that the fluctuation of AF^{2}DN on the three datasets caused by the change of training samples is relatively small, and the performance is always at a high level. It is worth noting that the classification accuracy of the GRSS-DFC-2007 dataset is lower than that of the other two, which may be caused by the high similarity of ground object categories and the small number of MSI bands. In addition, ExViT, CALC, and NCGLF^2 also show high accuracy in almost all sample numbers of all datasets, but we know that this is achieved at the expense of computational cost. In general, AF^{2}DN shows robustness under different numbers of training samples. Whether it is a small sample or a complex scene, our proposed method shows strong competitiveness.

E. Ablation Study

1) Ablation Experiments on Different Modules in AF^{2}DN: To validate the necessity of the proposed modules, we performed ablation studies on key components of AF^{2}DN. The MSFE module extracts multimodal features at an early stage and decomposes the data into four different frequency components through TFFD. The AMSF^3 module fuses features, while the AFMM further refines them and reduces redundancy. Table V summarizes the ablation results on three datasets.

We ablate the MSFE module and AFMM by completely removing them. The AMSF^3 module is ablated via a direct concatenation operation. For the four frequency components, we replaced their window inputs with the smallest available size. AF^{2}DN achieved the best performance when all modules were included. Removing MSFE decreased OA by 0.82%, 0.38%, and 0.49% across the datasets, with the largest drop on Houston2013, likely due to its rich edge information. Absence of HHF caused minimal reduction, as HHFs used a 1×1 window under our computational constraints. Omitting AMSF^3 resulted in the most substantial declines: 2.23%, 0.98%, and 0.88%, highlighting the critical role of same-frequency feature fusion. Finally, removing AFM reduced OA by 1.00%, 0.34%, and 0.46%, indicating that feature decomposition and fusion alone are insufficient—explicit frequency enhancement and global learning are also essential.

TABLE VI
ABLATION RESULTS OF THREE OPERATIONS OF MSFE ON THE HOUSTON2013 DATASET

Fusion Strategy	OA (%)	AA (%)	Kappa×100
Sum + Average + Max (Ours)	96.83	97.09	96.59
Sum + Average	96.55	96.92	96.39
Sum + Max	96.32	96.88	96.25
Average + Max	96.41	96.83	96.15
Sum only	96.25	96.73	96.07
Average only	96.18	96.71	95.88
Max only	96.15	96.65	95.73

TABLE VII
COMPARISON OF THE AMSF^3 MODULE WITH STANDARD FUSION METHODS

Fusion Method	Houston OA	Augsburg OA	GRSS DFC OA
Element-wise Sum	95.35	94.2	94.61
Concatenation	94.6	95.23	96.92
AMSF^3 (Ours)	96.83	96.21	97.8

2) Ablation Experiment of MSFE Fusion Strategy: To demonstrate the necessity of applying elementwise summation, elementwise averaging, and max pooling in the MSFE stage, we conducted ablation experiments on these three operations. As shown in Table VI, the experimental results clearly demonstrate that the proposed three-branch fusion strategy achieves optimal performance, with each component contributing positively to the overall outcome. The removal of any individual component led to a decrease in accuracy, which validates the rationality of our design.

3) Comparison of AMSF^3 With Standard Fusion Baselines: To directly compare AMSF^3 with a standard fusion baseline, we conduct fusion comparison experiments on AMSF^3. As shown in Table VII, AMSF^3 consistently outperforms these standard fusion strategies under the same conditions, typically delivering a 1%–3% improvement in OA. This is due to its adaptive weighting mechanism, which more effectively

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integrates co-frequency information across modalities and suppresses redundancy.

V. Conclusion

This article introduces the AF^{2}DN, a novel framework for hyperspectral and LiDAR data classification. The processing pipeline begins with the reduced-dimensional patches entering the MSFE module, which integrates multiscale information to generate multimodal shallow features. These features subsequently undergo frequency decomposition in the TFFD module, where window attention mechanisms extract multiple frequency components. The framework then employs the AMSF^3 module with its coarse-to-fine weight allocation strategy to enhance fusion completeness and focus on discriminative information. In the final stage, the AFMM refines the frequency feature representations to optimize their utilization in the model. Experimental results demonstrate that AF^{2}DN achieves competitive performance compared with state-of-the-art methods. Nevertheless, the computational complexity remains relatively high for certain datasets, suggesting potential areas for optimization. In the future, our research focus will be to simplify the model by reducing parameters and try to use some unsupervised methods to save costs and avoid annotation bias. In addition, we will further improve the efficiency of feature fusion so that the frequency-domain decomposition method can be more widely used in multimodal remote sensing data classification tasks.

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